

AstroBot

An Intelligent Conversational AI Assistant
with Multimodal Capabilities, RAG and Workflow Integration

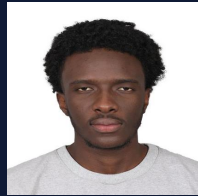
MFBP 402 — Graduation Project II

Computer Engineering · Faculty of Engineering
OSTIM Technical University



Tidiane Konaté

Architecture · Backend ·
Deployment



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Vision · Product · UX



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QA · Frontend · Testing

May 18 — 21, 2026 · Advisor: Assist. Prof. Dr. Yücel TEKİN



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01

Introduction & Vision

Why AstroBot — the problem, the goals, the users

PRESENTER

Sidi Mohamed Sall

02

Architecture & Design

System model, tech stack, data and security flow

PRESENTER

Tidiane Konaté

03

Implementation

Chat, image gen, RAG, OCR, voice, productivity

PRESENTER

Ahmed Essalem

04

Admin · Testing · Deployment

Dashboard, 66 tests, Docker stack, live MVP

PRESENTER

Saad Ibrahim Houssein

01

Introduction & Vision

Presented by Sidi Mohamed Sall

Why AstroBot — the problem, the goals, the users.

Fragmented tools

Chat, image gen, OCR, voice and document Q&A live in different apps. Users waste time switching contexts.

Paywalls everywhere

Most useful features are gated behind subscriptions or per-call fees, which puts students and small teams off-side.

Disposable conversations

Sessions are flat lists. No tags, no folders, no notebook context, no exportable record of what was discussed.

Black-box workflows

Multi-step prompt pipelines are either rebuilt from scratch every time, or invisible inside closed agents.

01

Unify modalities

Text, image, audio, OCR, document Q&A and code highlighting in a single chat surface.

02

Stay free for users

Combine free-tier APIs (HuggingFace, Mistral, Groq) and self-hosted models behind one wrapper.

03

Make sessions persistent

Folders, tags, notebooks, exports — give every conversation a memory and a paper trail.

04

Show our work

Workflows are saved, replayable and visible — including the n8n graph that drives the agent.

05

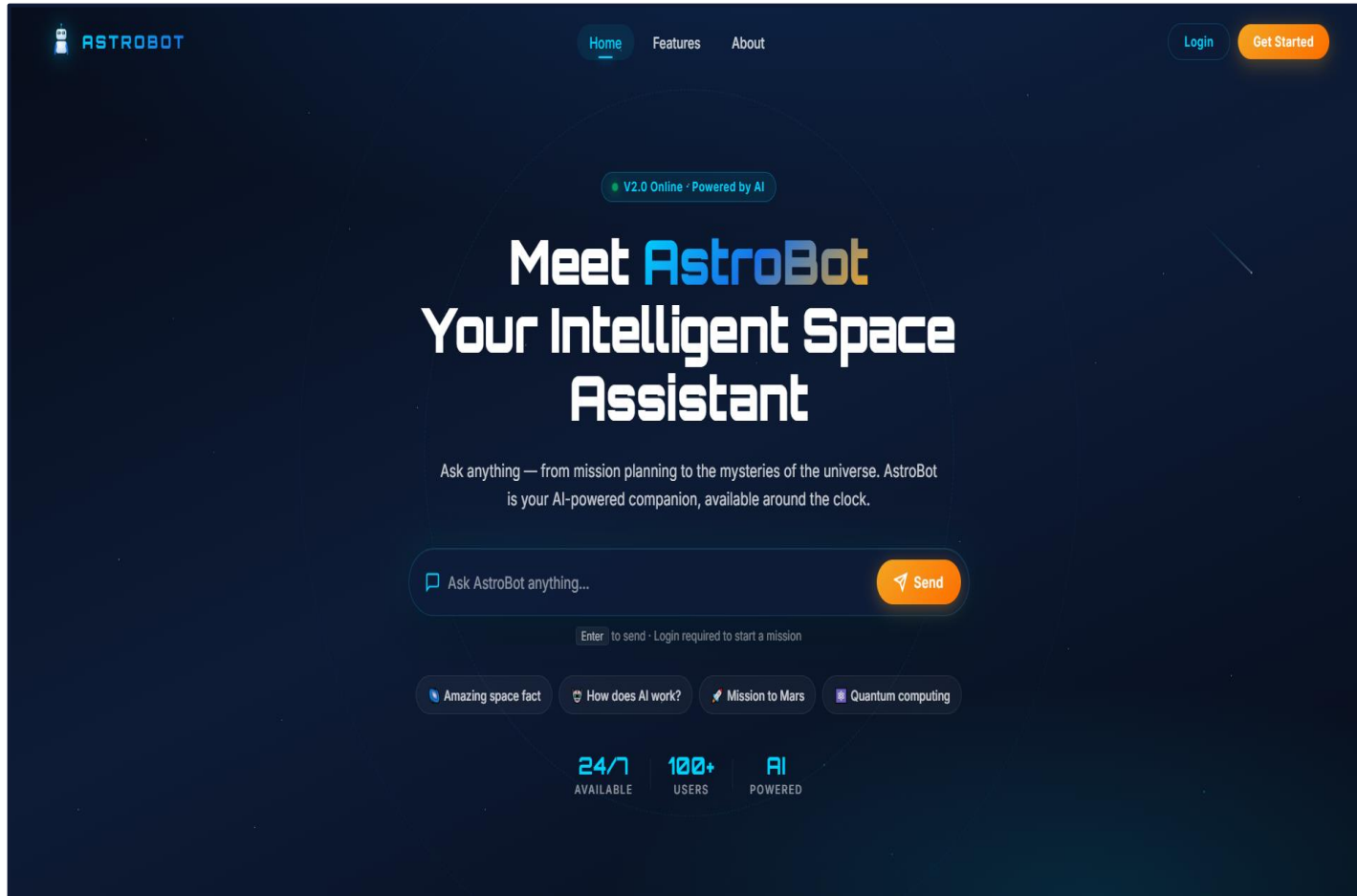
Ship a real MVP

Live, dockerised, on a real server, with admin tooling and a 66-test suite — not a prototype.

06

Ground on academic rigor

Defend every choice with measurements, diagrams and reproducible deployment scripts.



Public landing page — central chat bar, animated robot mascot.

Target users

- Students prepping reports & code
- Developers exploring multimodal AI
- Knowledge workers doing doc Q&A
- Anyone who wants one tab, not ten

Scope of the MVP

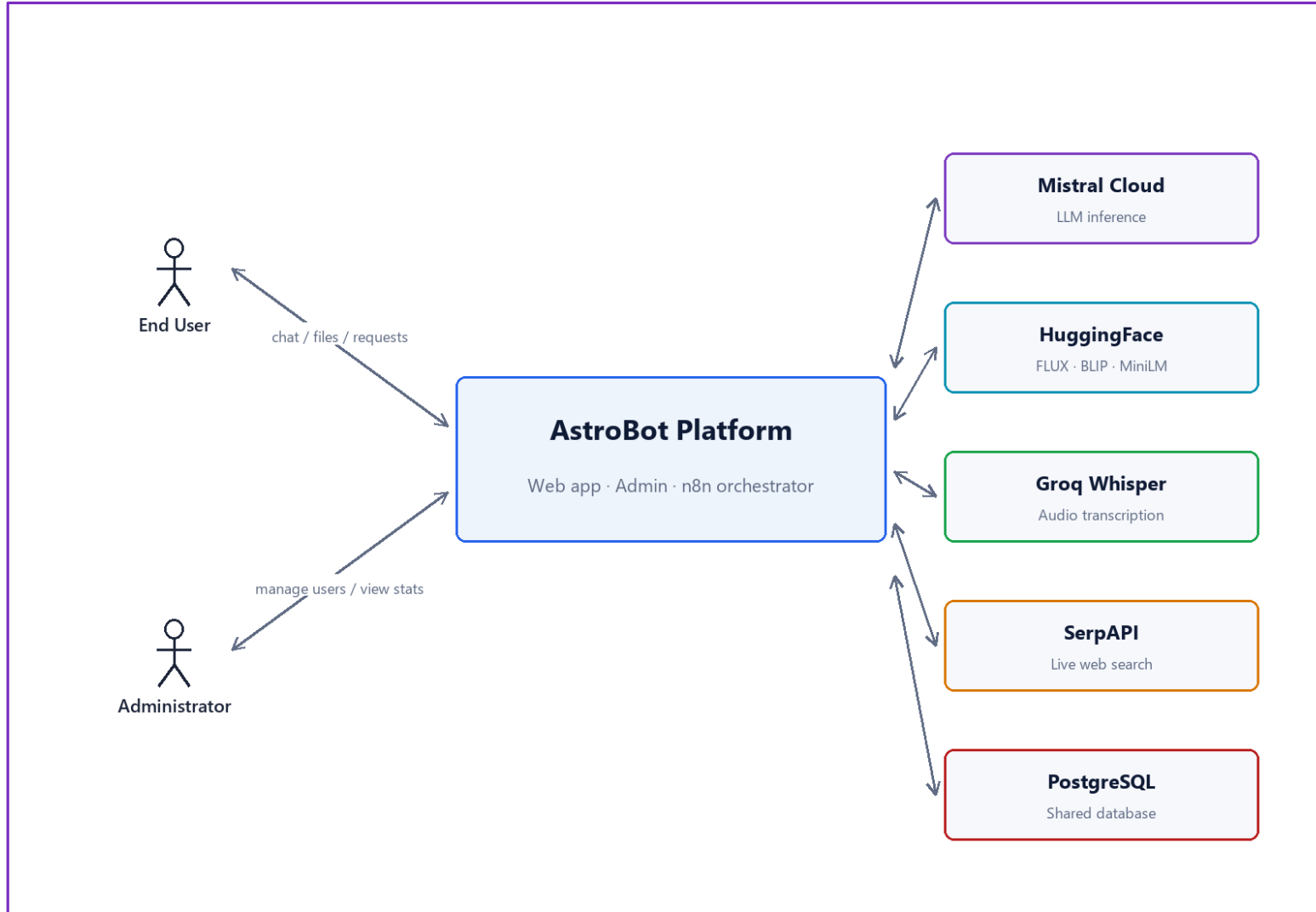
- Public + authenticated chat surface
- Admin dashboard with real metrics
- 12+ AI-powered features (see Part 3)
- Production deployment on a real VM

02

Architecture & Design

Presented by Tidiane Konaté

How the pieces fit together — services, data, security.



What this shows

AstroBot sits in the middle. Around it are the people and external services it talks to.

Actors

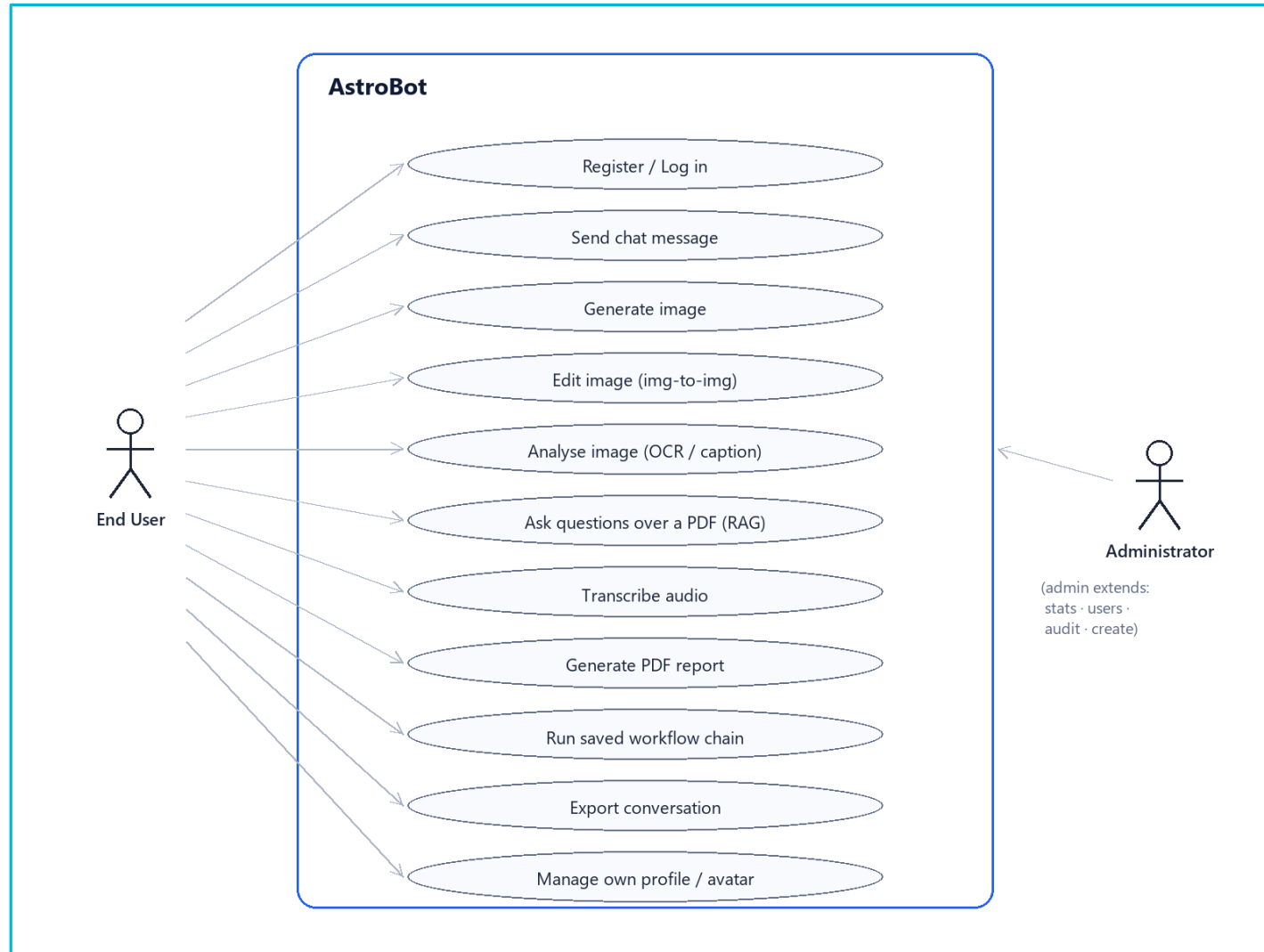
- End user — chats, uploads, listens to replies.
- Administrator — runs the dashboard, manages accounts and audits activity.

External services

- Mistral Cloud — the language model behind every reply.
- HuggingFace — image generation, captioning, embeddings.
- Groq Whisper — voice-to-text.
- SerpAPI — web search tool for the AI agent.

Boundary

Everything outside the central box is out of our control and reached over HTTPS only.



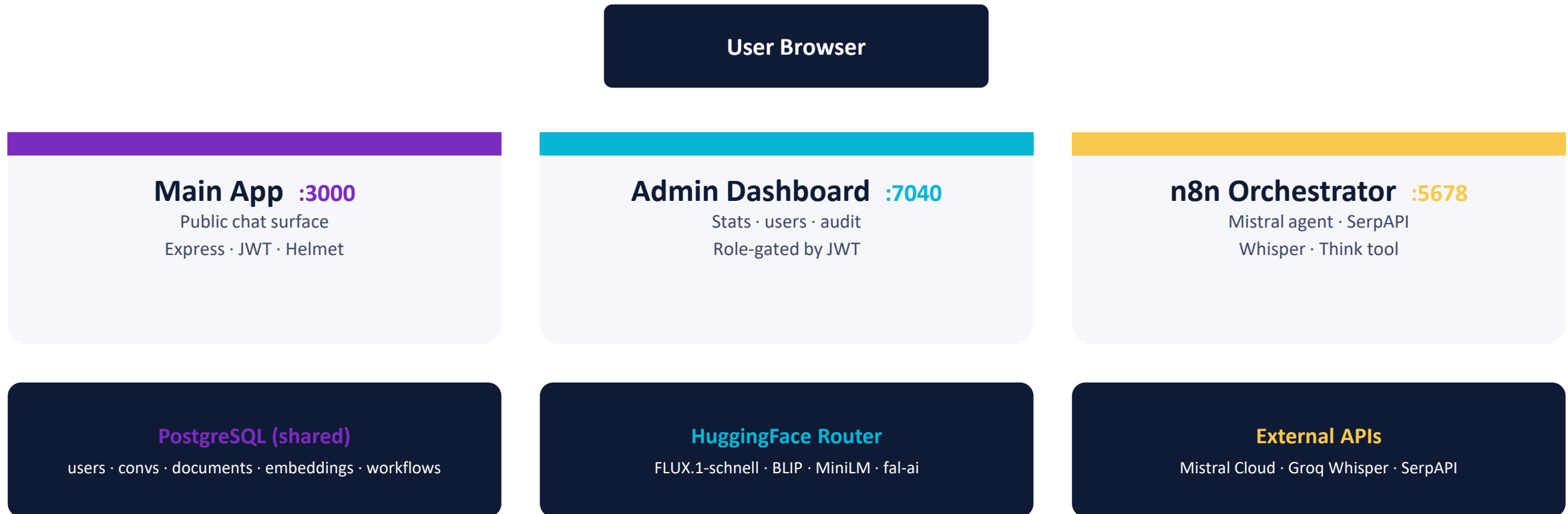
End user

- Register & log in (JWT)
- Chat in seven personas
- Generate or edit images
- Ask questions over a PDF (RAG or notebook)
- Speak a message (Whisper transcription)
- Read an image (OCR + BLIP captioning)
- Generate a PDF report on demand
- Save, tag, search and export sessions
- Compose multi-step workflow chains

Administrator

- Log in to the admin dashboard
- View KPIs and activity charts
- Search, suspend, reactivate, delete or create users
- Reset a user's password
- Inspect any user's conversation history

Every use case here also goes through the JWT auth + role gate.



Why this shape:

- Reuses an existing PostgreSQL container — no extra DB to operate.
- All AI traffic exits through n8n, so prompts and tools can be tuned without redeploying.
- Admin lives in its own service so a public-facing exploit can never reach the dashboard process.

Frontend

- Vanilla HTML / CSS / JavaScript
- KaTeX · highlight.js · Mermaid
- Cropper.js · Chart.js (admin)
- No framework — fast & maintainable

Backend

- Node.js 18 · Express 4
- JWT · bcrypt 12 rounds
- Helmet · express-rate-limit
- pg, pdfkit, tesseract.js, pdf-parse

AI / LLM

- Mistral medium-latest (via n8n agent)
- HuggingFace FLUX.1-schnell · BLIP
- MiniLM-L6-v2 embeddings
- Groq Whisper-large-v3-turbo

Infrastructure

- Docker · Docker Compose
- Shared PostgreSQL container
- n8n network (existing)
- Python deploy.py (paramiko)

users

id · email · password_hash
is_admin · status · avatar · last_login

conversations

id · user_id · message · response
session_id · session_tag · image_url
pdf_url · pdf_filename · attachment

documents

id · user_id · name · mime
size · text_content · created_at

document_chunks

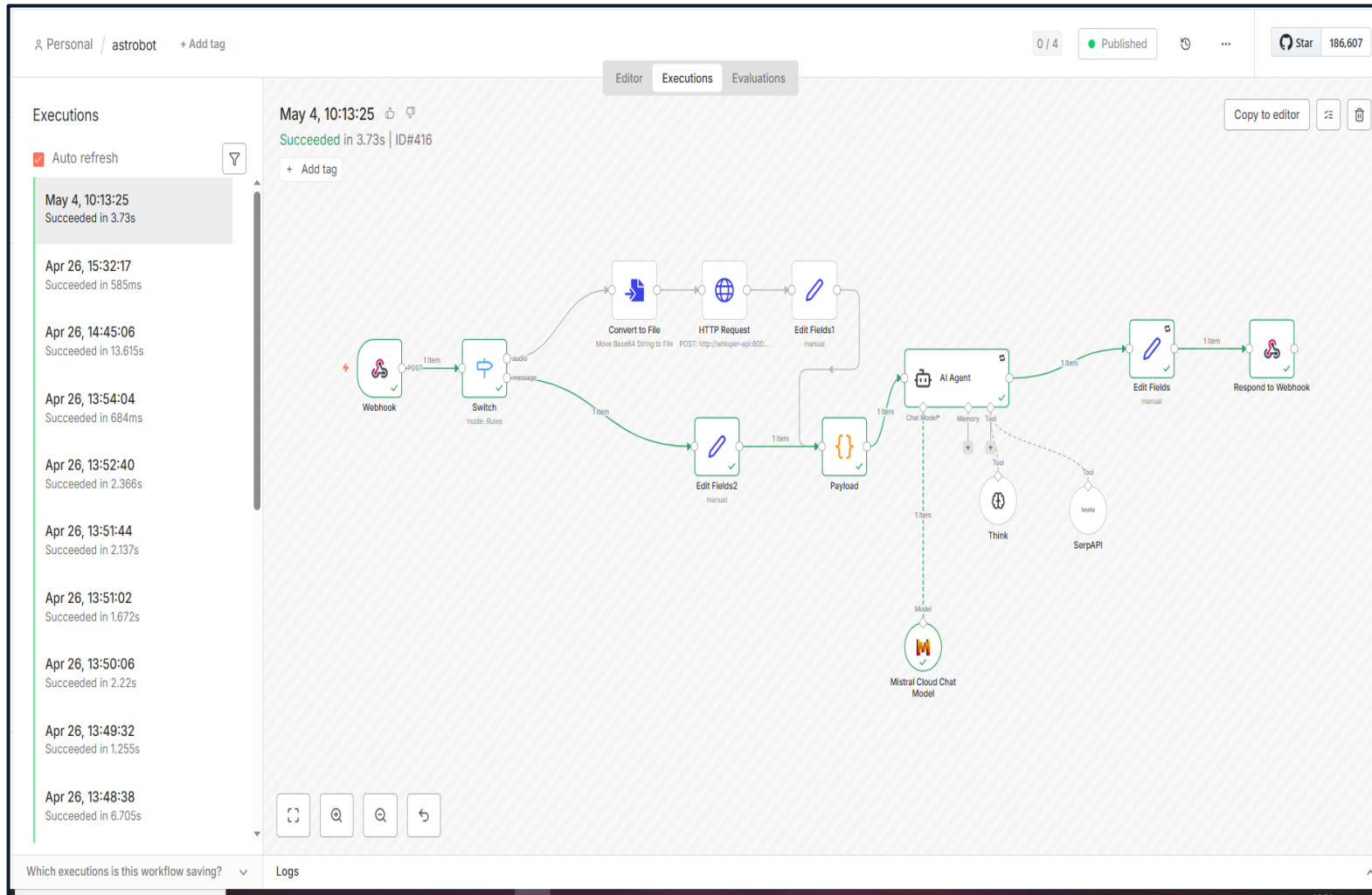
document_id · chunk_index
chunk_text · embedding (TEXT)

notebook_bindings

user_id · session_id · document_id
mode (notebook | rag)

workflows

id · user_id · name
steps (JSON)



Pipeline

- Webhook** — POST receiver
- Switch** — audio vs text branch
- Whisper** — audio transcription
- Payload** — history + persona enrichment
- AI Agent** — Mistral medium · 20-iter cap
- SerpAPI** — live web search (max 2 calls)
- Think** — scratchpad reasoning
- Edit Fields** — extracts JSON `response`
- Respond** — back to backend

01

Authentication

- JWT (7-day expiry) signed with rotating secret
- bcrypt password hashing (12 rounds)
- Admin gate via `is\_admin` JWT claim

02

Network surface

- Helmet HTTP security headers
- CORS allow-list per origin
- No cache on static assets

03

Rate limits

- Auth · 20 / 15 min
- Chat · 30 / min
- Global · 200 / 15 min · Admin · 500 / 15 min

04

Operational hygiene

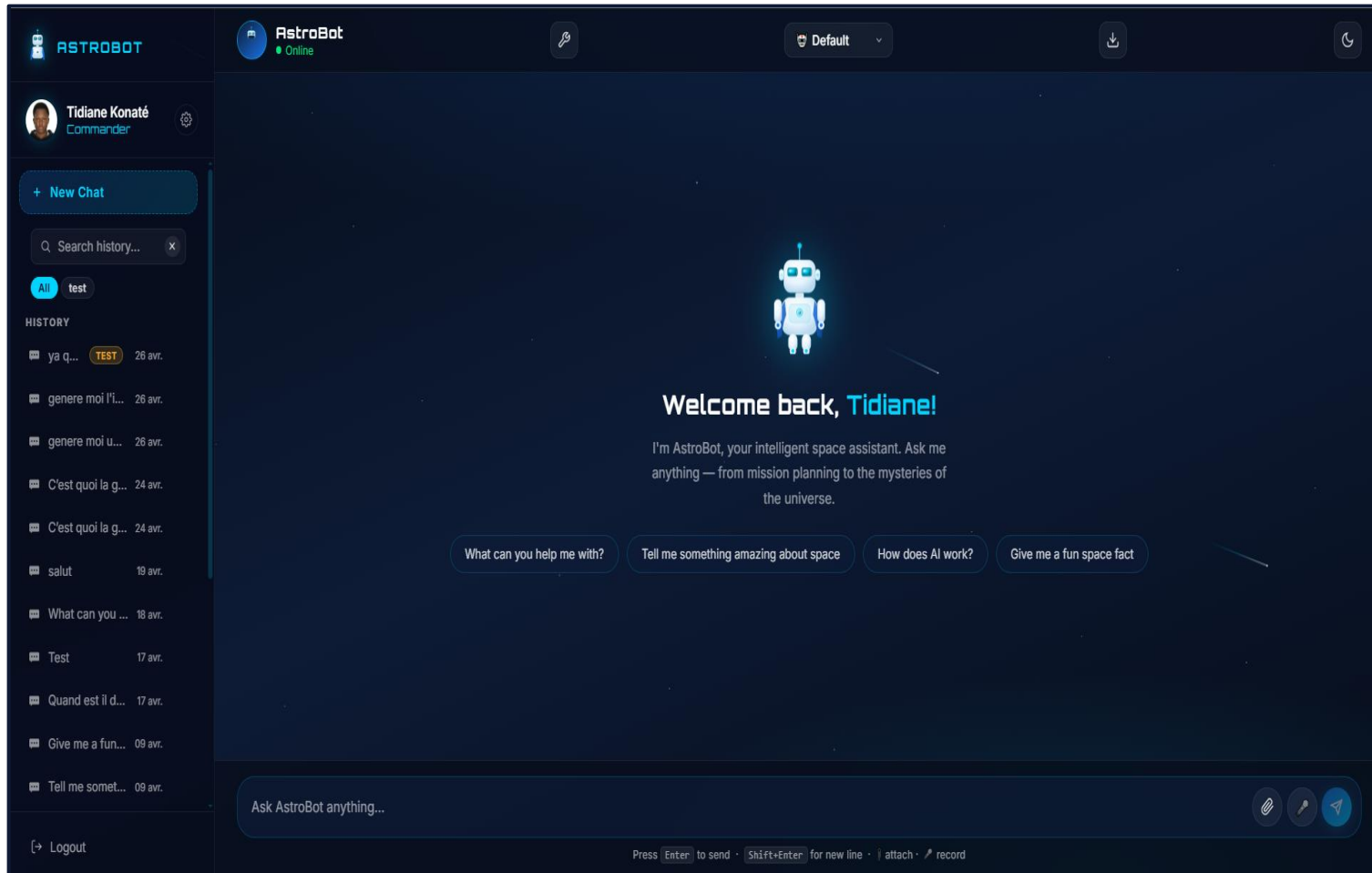
- Secrets in .env files (gitignored)
- File uploads capped 10 MB · 5 attachments
- Idempotent migrations + admin re-seed at boot

03

Implementation

Presented by Ahmed Essalem

The features — chat, multimedia, RAG, productivity.



Authenticated chat — welcome screen with quick-start prompts.

Mistral medium via n8n

15-message context window per session, persona directives prepended.

Word-by-word reveal

Animated streaming UX without server-side streaming complexity.

Markdown · Math · Code · Diagrams

GitHub-flavored Markdown, KaTeX, highlight.js, Mermaid all live in the same bubble.

Multi-language

Auto-detect FR / EN / ES / DE / AR / ZH / JA / KO / RU / EL.

7 personas

Default · Formal · Casual · Teacher · Developer · Poet · Scientist.

Markdown

Lists, headings, emphasis, blockquotes, links — sanitised on output.

LaTeX math

$E=mc^2$ inline, $\int_0^\infty$ block, KaTeX renders client-side.

Code highlighting

highlight.js detects 180+ languages; one-click copy buttons.

Mermaid diagrams

Flow charts, sequence diagrams, ER models, Gantt — straight from text.

Markdown rendering with structured lists, emphasis and code.

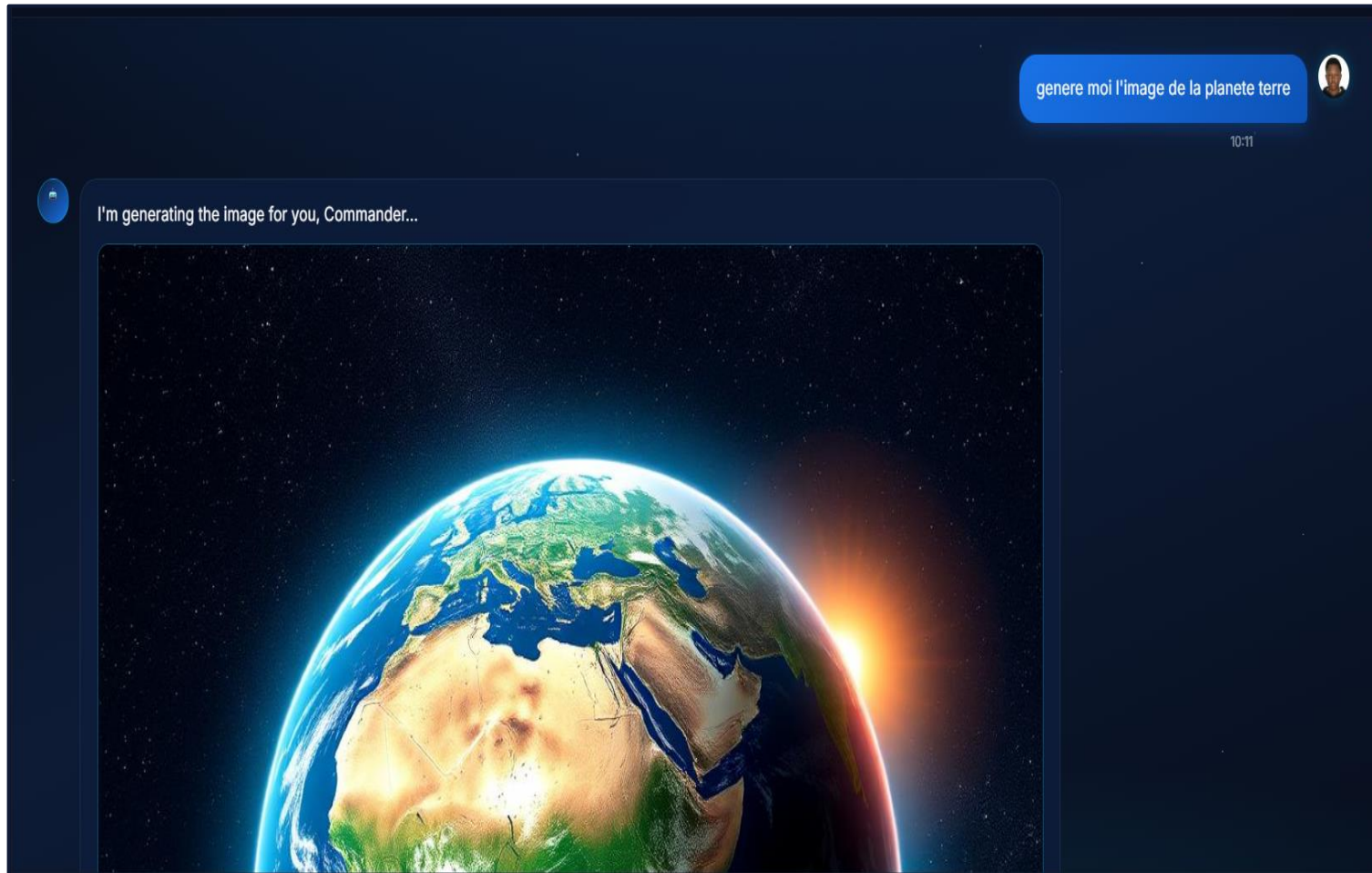


Image generation example — prompt-to-render in a single bubble.

FLUX.1-schnell

HuggingFace router endpoint — 4-step diffusion model, free tier.

fal-ai fallback

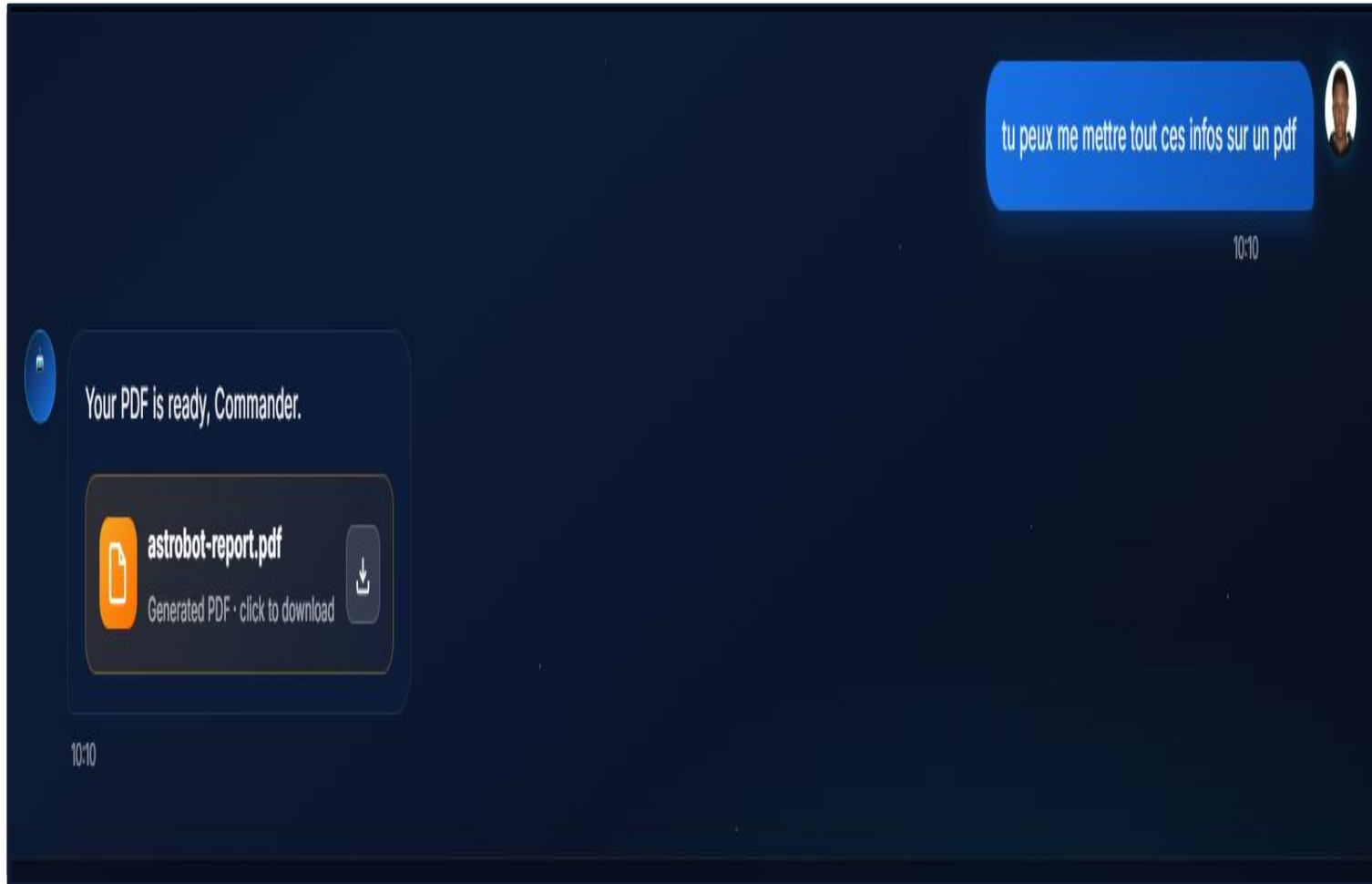
Automatic retry path when HF rate-limits or times out.

Image-to-image

Style transfer (anime, oil painting, etc.) via FLUX dev img2img.

Inline rendering

Image is stored as data URL on the conversation row — survives reloads.



A generated PDF surfaced as a download card inside the chat.

PDFKit

Server-side composition with title, body, sections and styled emphasis.

Heuristic action triggers

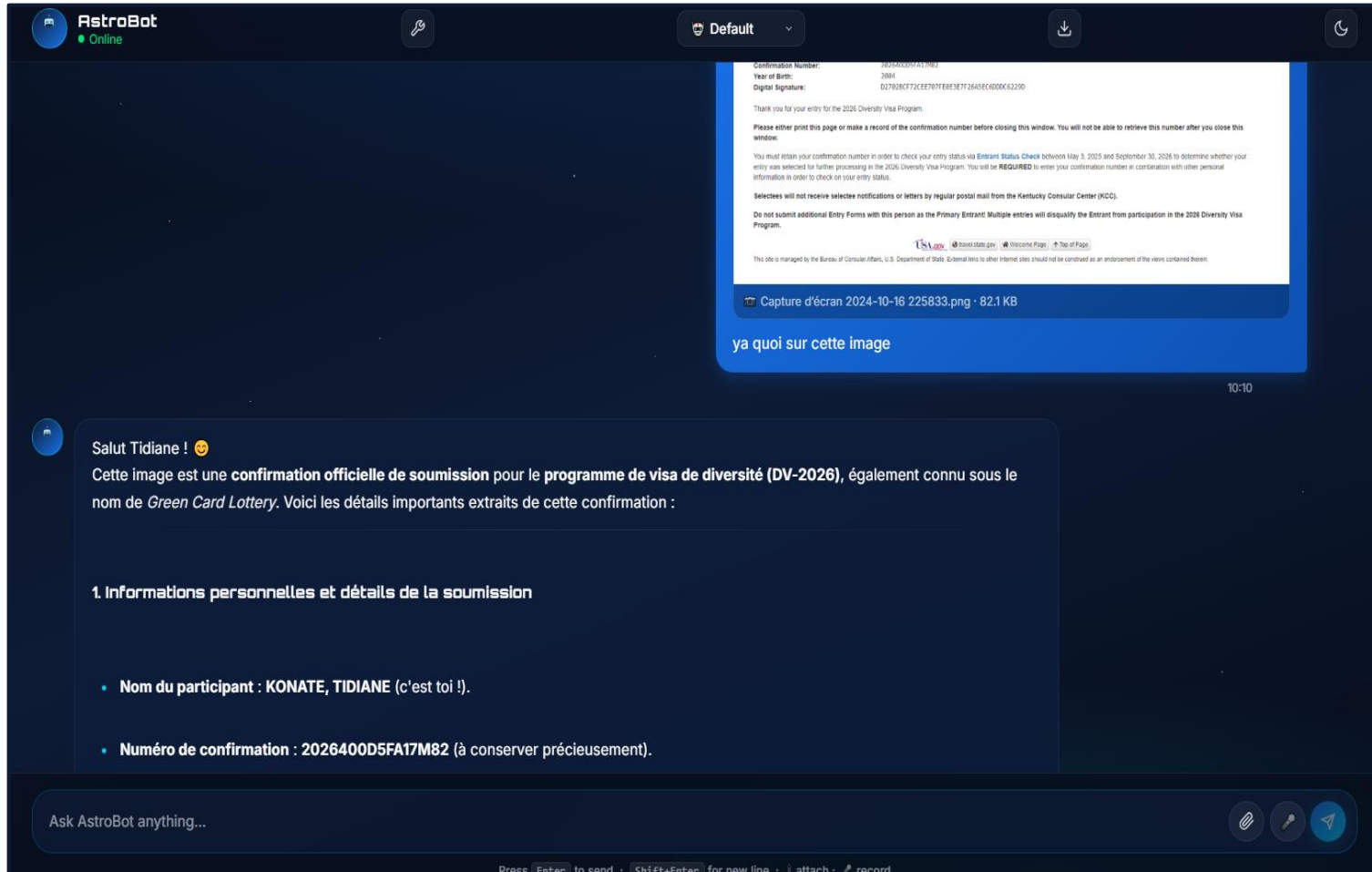
Backend re-parses the agent reply — if it acknowledges the report, we force ``generate_pdf``.

Inline preview card

Filename, size badge, one-click download from inside the bubble.

Persisted

Stored as data URL in the conversation row for re-download anytime.



OCR + reasoning — the AI explains a Green Card lottery confirmation.

Tesseract.js

On-server OCR · English + French · embedded as message context.

BLIP captioning

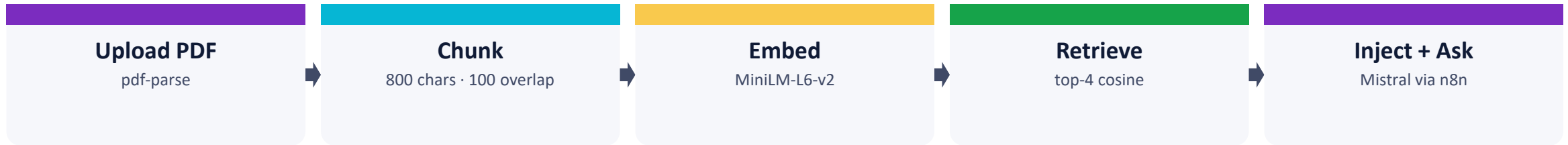
HuggingFace BLIP visualizes the image when there's no text.

Combined into context

Both signals are concatenated to the prompt before reaching Mistral.

Use cases

Forms, screenshots, error messages, scanned documents, ID cards.



Notebook mode

First 6,000 characters of the bound document are **always** in context. Best for short PDFs or when you want the AI to discuss the whole document.

RAG mode

On every question we embed the prompt with MiniLM and pull the top-4 most similar chunks (cosine similarity) into context. Scales to long documents.

Why we kept both:

- Notebook is dependable for thesis-sized PDFs (<10 pages).
- RAG is the only thing that makes 200-page books answerable in real time.

01

Workflow chains

Save multi-step prompt sequences and replay them on a single click.

02

Folders & tags

Color-coded session tags · search across full conversation history.

03

Exports

Conversation as PDF · Markdown · JSON — three clicks each.

04

YouTube summarizer

Paste a URL, get a transcript-grounded summary inside the chat.

05

Audio transcription

Browser recording → Groq Whisper-large-v3-turbo → typed reply.

06

Profile + cropper

Built-in image cropper (rotate · zoom · flip) for avatars.

04

Admin · Testing · Deployment

Presented by Saad Ibrahim Houssein

How we operate, audit and ship the product.



Admin dashboard — KPIs, daily message chart, status doughnut.

Real-time KPIs

Users · messages · sessions · tokens · images · PDFs · documents.

Charts

Daily activity (30 days) and user-status doughnut, both Chart.js.

Recent signups & top users

Quick triage panels visible above the fold.

Hard role gate

Login refuses any account that is not ``is_admin = TRUE``.

ASTROBOT
ADMIN CONSOLE

Dashboard
Users
Settings
Open AstroBot

Tidiane Konaté
ADMINISTRATOR
Logout

Users

Manage commanders, status and permissions

Search by name or email... + New user

USER	EMAIL	STATUS	ROLE	SESSIONS	MESSAGES	LAST LOGIN	JOINED	
test test ID #8	test@test.com	ACTIVE	USER	0	0	—	26 Apr 2026, 14:15	
Coumba MAGUIRAGA ID #7	coumbaayoubam@gmail.com	ACTIVE	USER	1	2	—	25 Apr 2026, 03:34	
Sidi Mohamed Sall ID #6	sidimohamedsall5@gmail.com	ACTIVE	USER	1	5	—	25 Apr 2026, 03:34	
Mamadou Keita ID #5	mkeita5912@gmail.com	ACTIVE	USER	2	6	—	24 Apr 2026, 14:21	
saad ibrahim ID #4	saadgodin12@gmail.com	ACTIVE	USER	4	14	—	18 Apr 2026, 20:04	
Modi Niang ID #3	modiniang1975@gmail.com	ACTIVE	USER	1	1	—	2 Apr 2026, 00:42	
Tidiane Konaté ID #2 - You	tkonate788@gmail.com	ACTIVE	ADMIN	20	50	4 May 2026, 10:20	12 Mar 2026, 15:54	

User management table with status, role and recent activity columns.

Searchable table

Status · role · sessions · messages · last login · joined date.

Lifecycle actions

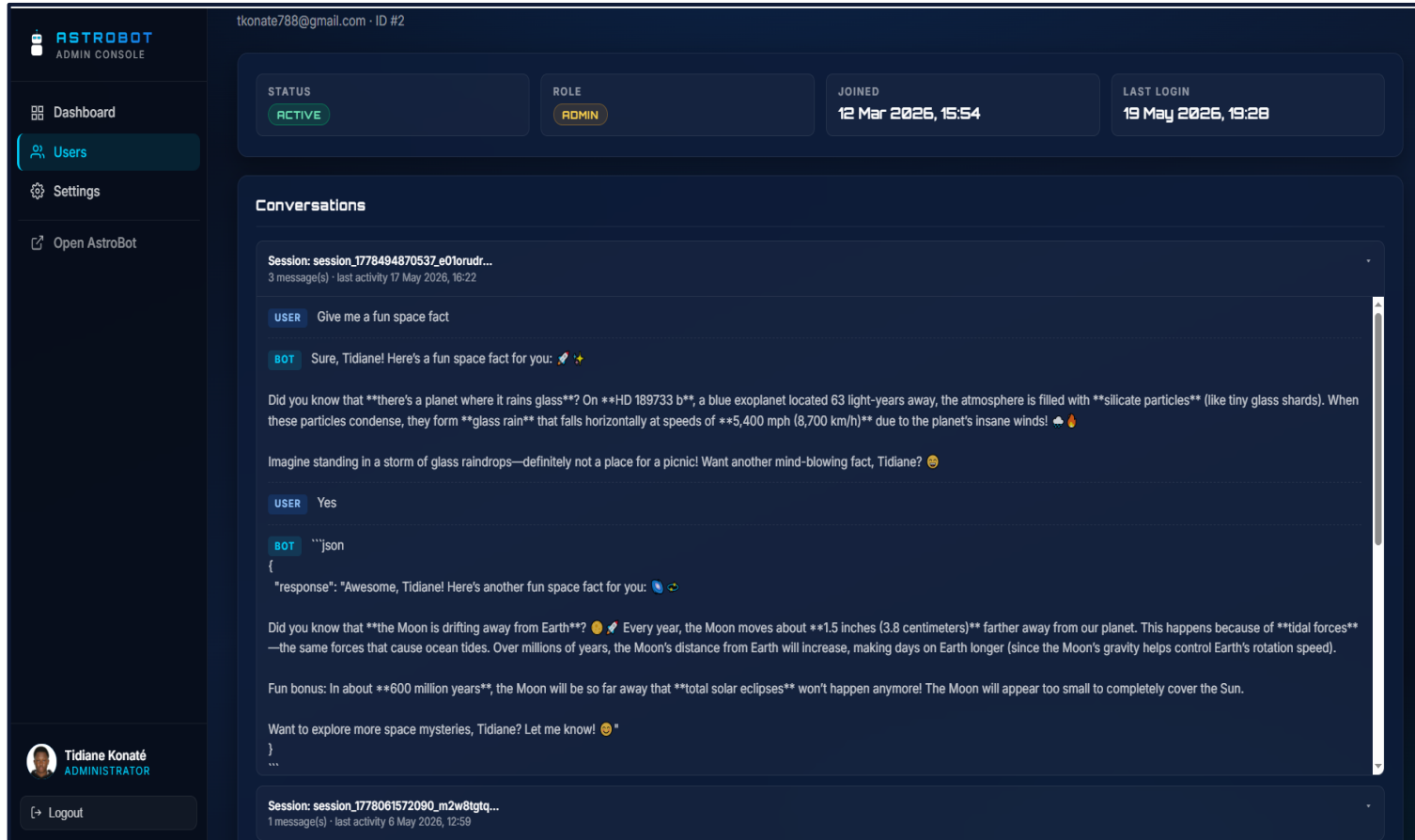
Suspend / activate · reset password · delete user (self-delete blocked).

Account creation

Admins create regular or admin accounts directly from the UI.

Soft 403 responses

Non-admin tokens are rejected before any state is read.



The screenshot displays the AstroBot Admin Console interface. On the left is a sidebar with navigation links: Dashboard, Users (selected), Settings, and Open AstroBot. The main content area shows the profile of user 'tkonate788@gmail.com - ID #2'. At the top, there are four status boxes: STATUS (ACTIVE), ROLE (ADMIN), JOINED (12 Mar 2026, 15:54), and LAST LOGIN (19 May 2026, 19:28). Below this is a 'Conversations' section with a scrollable list of message sessions. The first session, 'session_1778494870537_e01onudr...', shows a user asking for a fun space fact and the bot providing a detailed response about 'glass rain' on an exoplanet. The second session, 'session_1778061572090_m2w8tgtq...', shows a single message. At the bottom left, there is a user profile for 'Tidiane Konaté' (ADMINISTRATOR) and a 'Logout' button.

Profile metadata

Email · joined · status · role · counts at a glance.

Sessions list

Each session previewed with its first message and tag.

Drill-down

Open any session to see the full message + response history.

Compliance-ready

Built so we can answer a 'what did this user ask?' request in seconds.

User detail — full conversation history grouped by session.

Health Main + admin /health endpoints reachable

Auth Login · wrong password · missing token · role check

Main app Profile · avatar set + clear · password change

Sessions Tags · history · session detail · session delete

Features Documents · RAG bind/unbind · workflows CRUD

Static assets HTML/CSS/JS served with no-cache headers

Admin CRUD Stats · users list · create · update · suspend · reset · delete

Chat flow Round-trip via n8n → Mistral → response

Exports PDF · Markdown · JSON exports validate

Validation Bad inputs return clean 400/422 — never 500

deploy.py — five remote commands

1. `tar -cz frontend backend admin docker`
2. SFTP upload to /tmp on the VM
3. `ssh: tar -xz into ~/astrobot`
4. `docker compose up -d --build`
5. healthcheck verifies both services

Live containers

`astrobot:2.0.0` :3000 *main app*
`astrobot-admin:1.0.0` :7040 *admin dashboard*
`postgre_container` :5432 *shared (read/write own DB only)*
`n8n` :5678 *external · do not modify*

Try it right now

Web app	•	<code>http://76.13.62.195:3000</code>
Admin	•	<code>http://76.13.62.195:7040</code>
Source	•	<code>github.com/tkonate788/astrobot</code>

Lessons learned

Cache-Control: no-store on iOS

Mobile Safari held onto stale CSS for hours; killed it at the express layer.

Heuristic action triggers

LLMs occasionally agree to a PDF without flagging it — backend re-parses replies.

Idempotent migrations

ALTER TABLE ... ADD COLUMN IF NOT EXISTS lets the app self-heal on every boot.

Single deploy script

Beats `git pull` over SSH — no half-deployed state on partial failures.

What's next

Mobile-first PWA

Service worker · push notifications · standalone app feel.

Voice mode

Two-way audio loop — Whisper in, Web Speech out.

More languages

Tesseract Arabic + Cyrillic; persona auto-tuning per language.

Self-hosted Whisper

Already in n8n graph — extend it to all audio paths.

Plug-in marketplace

Workflows shared and rated across users.

12+

AI-powered features, all in one place
— chat, image, OCR, RAG, voice, PDF,
workflows.

66

Integration tests covering auth, chat,
admin and features; runnable in one
command.

3

Dockerised services on a single shared
network — reproducible from a fresh
VM.

4

Engineers, four roles, four months —
no feature got dropped along the way.

AstroBot is not a demo — it is a working product.

Every feature shown today runs against the live VM at <http://76.13.62.195:3000> — feel free to log in during Q&A.

Thank you for your attention!

QUESTIONS WELCOME

Try AstroBot live

Web app • <http://76.13.62.195:3000>
Admin • <http://76.13.62.195:7040>
GitHub • github.com/tkonate788/astrobot



Ostim, 100. Yıl Blv 55/F, Yenimahalle/Ankara



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